

Required:

Introduction (Aron and Enzo)

Mathematical Model (Complete- Krishiv)

Code/program (Alex and Abi)

Analysis (Manu, Lucas, Gabi)

Other methods/ways to improve (Alex, Krishiv, Diego)

Conclusion (Marco and Gabi)

Video

USE TIMES NEW ROMAN SIZE 12 FOR ALL MAIN BODY TEXTS, AND HEADING 1 FOR ANY TITLES, AND HEADING 2 FOR SUBTITLES

Introduction (Enzo and Aron)

Introduction of variables and model

To model the theoretical patient capacity at a hospital, we have elected to use the following variables:

$d = \text{number of doctors}$

$n = \text{number of nurses}$

$e = \text{number of equipment (bundled)}$

$b = \text{number of beds}$

$p = \text{number of patients}$

Based on these variables, we have developed the following constraints:

$$\text{Base constraint: } P \leq \left(\frac{d}{d_p}, \frac{n}{n_p}, \frac{e}{e_p}, \frac{b}{b_p} \right)$$

where x_p refers to the quantity of variables per patient

$$\text{Budget Constraint: } C_d \cdot d + C_n \cdot n + C_e \cdot e + C_b \cdot b \leq B$$

where C_x refers to the cost of variable x and B refers to overall hospital budget

$$\text{Minimum Care Quality: } n \geq p \cdot K$$

where K refers to the minimum required amount of nurses per patient

$$\text{Area Constraint: } A_e \cdot e + A_b \cdot b \leq A_T$$

where A_x refers to the area taken by variable x and A_T refers to total area available

$$\text{Non - negativity: } (d, n, e, b, p) \geq 0$$

Development and manipulation of the model

$$f = \max(p)$$

Since p is being maximized, we can re-write the base constraints in the form:

$$d = d_p \cdot p, \quad n = n_p \cdot p, \quad e = e_p \cdot p, \quad b = b_p \cdot p$$

We can substitute these values into the budget constraint:

$$C_d \cdot (d_p \cdot p) + C_n \cdot (n_p \cdot p) + C_e \cdot (e_p \cdot p) + C_b \cdot (b_p \cdot p) \leq B$$

We can re-arrange the equation to isolate p :

$$\begin{aligned} p(C_d \cdot d_p + C_n \cdot n_p + C_e \cdot e_p + C_b \cdot b_p) &\leq B \\ p &\leq \frac{B}{(C_d \cdot d_p + C_n \cdot n_p + C_e \cdot e_p + C_b \cdot b_p)} \end{aligned}$$

We can repeat the steps to write the area constraint in terms of p :

$$\begin{aligned} A_e \cdot (e_p \cdot p) + A_b \cdot (b_p \cdot p) &\leq A_T \\ p(A_e \cdot e_p + A_b \cdot b_p) &\leq A_T \\ p &\leq \frac{A_T}{(A_e \cdot e_p + A_b \cdot b_p)} \end{aligned}$$

Hence, we can write our final model as:

$$p \leq \min \left(\frac{B}{(C_d \cdot d_p + C_n \cdot n_p + C_e \cdot e_p + C_b \cdot b_p)}, \frac{A_T}{(A_e \cdot e_p + A_b \cdot b_p)} \right)$$

With the feasibility constraint:

$$\begin{aligned} n &\geq p \cdot K \\ n_p \cdot p &\geq p \cdot K \\ n_p &\geq K \end{aligned}$$

And the non-negativity constraints:

$$(d, n, e, b, p) \geq 0$$

This model, alongside its related constraints, can be developed into a model on Python that can solve for the maximum capacity a hospital can maintain with a set of given variables.

Code (Alex and Abi)

Analysis (Manu, Lucas, Gabriel)

Areas to improve (Alex, Krishiv, Diego)

Conclusion (Gabriel and Marco)